

## Project Initialization and Planning Phase

|               |                                                   |
|---------------|---------------------------------------------------|
| Date          | 7 July 2026                                       |
| Team ID       |                                                   |
| Project Title | EduGenie Google Gemini Powered Learning Assistant |
| Maximum Marks | 5 Marks                                           |

### Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

| Project Overview  |                                                                                                                                                                                                                    |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective is to revolutionize the learning assistance process by implementing advanced generative AI techniques, ensuring faster concept synthesis and more accurate academic assessments.             |
| Scope             | The project comprehensively assesses and enhances the traditional study process, incorporating local and cloud machine learning models for a more robust and hardware-efficient system.                            |
| Problem Statement |                                                                                                                                                                                                                    |
| Description       | Addressing gaps, static content, and text overload in the current learning ecosystem, as rigid or un-yielding educational platforms adversely affect study efficiency and student satisfaction.                    |
| Impact            | Solving these issues will result in improved learning efficiency, reduced cognitive fatigue, and an overall enhancement in the self-evaluation process, contributing to student satisfaction and academic success. |
| Proposed Solution |                                                                                                                                                                                                                    |
| Approach          | Employing generative AI techniques to analyze queries and predict optimized learning paths, creating a dynamic and adaptable educational support system.                                                           |

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| Key Features | Implementation of a hybrid machine learning-based educational assessment and summarization pipeline using local and cloud models. |
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Resource Requirements

| Resource Type           | Description                             | Specification/Allocation                                                                                                         |
|-------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Hardware                |                                         |                                                                                                                                  |
| Computing Resources     | CPU/GPU specifications, number of cores | Optimized for multi-core processors (e.g., Apple M1, Intel i3/i5 or higher) with support for lightweight AI inference execution. |
| Memory                  | RAM specifications                      | 4 GB minimum (8 GB recommended for smoother local model loading).                                                                |
| Storage                 | Disk space for data, models, and logs   | 10 GB free disk space (to store local LaMini model weights, application files, and session databases).                           |
| Software                |                                         |                                                                                                                                  |
| Frameworks              | Python frameworks                       | Flask                                                                                                                            |
| Libraries               | Additional libraries                    | FastAPI, Uvicorn (ASGI web server platform).                                                                                     |
| Development Environment | IDE                                     | Google Generative AI SDK, Transformers, Torch, Jinja2 template utilities.                                                        |
| Data                    |                                         |                                                                                                                                  |
| Data                    | Source, size, format                    | Visual Studio Code, PyCharm,                                                                                                     |

|  |  |                                                         |
|--|--|---------------------------------------------------------|
|  |  | or any standard text editor<br>supporting Python 3.10+. |
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